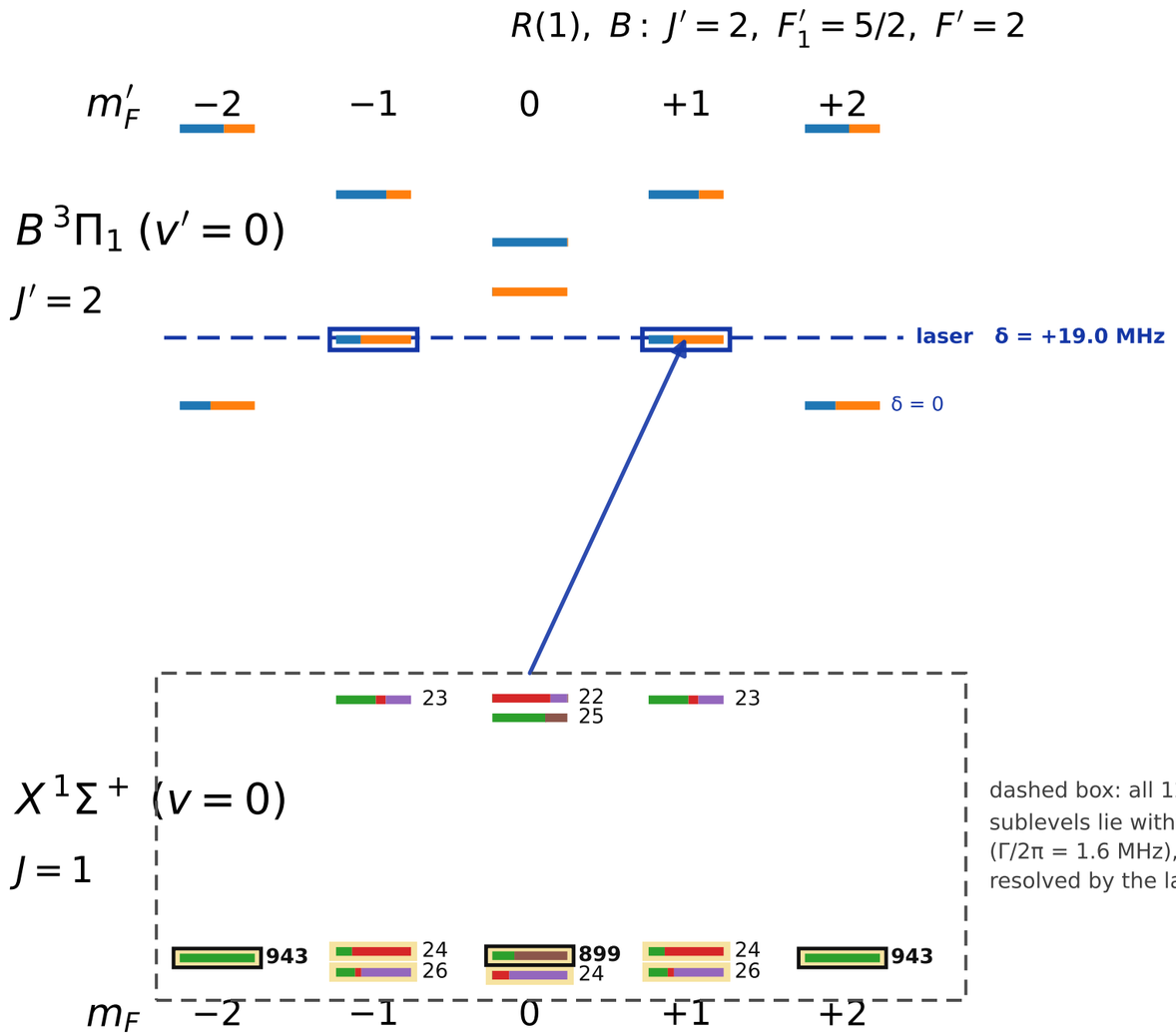
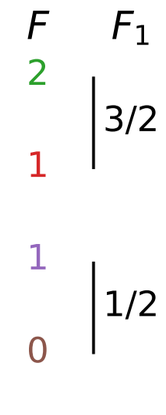
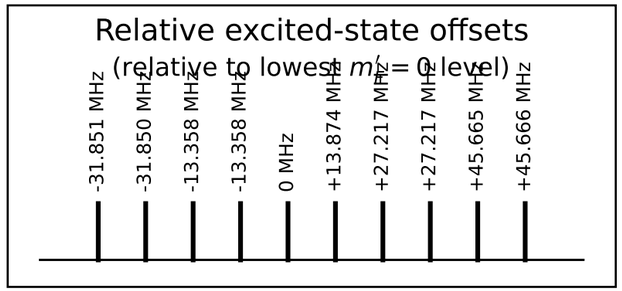
CU1 | R(1) $F_1' = 5/2$, $F' = 2$ | E = 150 V/cm | laser at +19.0 MHz



dashed box: all 12 J=1 sublevels lie within 2.4 MHz (Γ/2π = 1.6 MHz), not resolved by the laser





Opposite-parity-parent separation at 150 V/cm

$m_F' = -2$: 77.516 MHz
$m_F' = -1$: 40.575 MHz
$m_F' = 0$: 13.874 MHz
$m_F' = +1$: 40.575 MHz
$m_F' = +2$: 77.516 MHz

Zero-field parity splitting

13.906 MHz

Parity mixing at 150 V/cm (lower level)

$m_F' = -2$: 41.0/59.0
$m_F' = -1$: 32.9/67.1
$m_F' = 0$: 0.0/100.0
$m_F' = +1$: 32.9/67.1
$m_F' = +2$: 41.0/59.0

zero-field $P' = -1$ character
zero-field $P' = +1$ character

Ground-state zero-field hyperfine character

$F_1 = 3/2,\ F = 2$
$F_1 = 3/2,\ F = 1$
$F_1 = 1/2,\ F = 1$
$F_1 = 1/2,\ F = 0$

Dashed lines and arrows: CU1 laser, single frequency, no phase modulation. Drawn for the main ground level |F1=3/2, F=2, mF=+1>; delta = detuning from its resonance with |F1'=5/2, F'=2, mF'=+2, P'=+> (marked delta = 0).
For the other ground levels the lines move by -2.4 to +0.0 MHz. Coloured outline and arrow: sublevels a line reaches, i.e. within 4 Gamma/2pi = 6.4 MHz and with addressed-parity character > 5 % (line minus level): laser -> m'=±1 (P'+) +0.5 MHz.
Number right of each J=1 level: population left in J=1 after CU1 for a molecule starting in that level, in units of 1e-4 (80 mW, delta = +19.0 MHz, 4.8 cm beam, vx = 0, y0 = 0). Average over the 12 levels: 250.3e-4.
Bold black outline: residual more than twice that average. Yellow band: SPB-related level (J=1, |m_J|=1 content > 0.5; SPB transfers into these).
Single trajectory on the beam axis, no transverse velocity; 13 passes, 10 % loss per pass, X/Z switching at 0.5 MHz.
+19.0 MHz is the best detuning without EOM for a Gaussian vx distribution of 1.8 m/s FWHM (truncated at 1 m/s): averaged over vx and |y0| <= 15 mm it gives 95.2 %, against 93.1 % at +20.2 MHz, although on the axis at vx = 0 it is worse (96.36 % against 99.965 %).